Special Theory of Relativity

Unit 2: Relativity and Electromagnetism

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Reference books

1. D. J. Griffiths, *Introduction to Electrodynamics*, 4th ed. (Pearson, 2013), Ch. 12 (and §10.3)
2. J. R. Taylor, *Classical Mechanics* (University Science Books, 2005), §15.18

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Summary of Unit 2 **13**

1 Special relativity and electromagnetism: an introduction

Unit 1 told the story from the side of mechanics: Newton's laws had to be changed so that they obey the same transformation (Lorentz) as Maxwell's equations. Griffiths makes the striking point that *classical electrodynamics was already consistent with special relativity*: Maxwell's equations and the Lorentz force law $\mathbf{F} = q(\mathbf{E} + \mathbf{u} \times \mathbf{B})$ are valid in every inertial frame. What one observer calls an electric effect another may call magnetic, but the particle motions they predict are identical. In this unit we do not change electrodynamics at all – we rewrite it in a language that makes its relativistic character obvious.

1.1 Einstein's starting point: the magnet and the loop

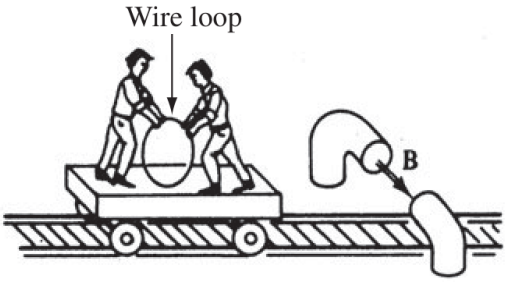


Figure 1: A wire loop on a cart moves past a stationary magnet (or the magnet moves past a stationary loop).
Source: D. J. Griffiths, Introduction to Electrodynamics*, Fig. 12.1*

- *Loop moves, magnet at rest:* charges in the moving wire feel a **magnetic** force $q\mathbf{u} \times \mathbf{B}$; this drives the current ("motional emf").
- *Magnet moves, loop at rest:* the charges are at rest, so there is no magnetic force; but the changing $\mathbf{B}$ induces an **electric** field (Faraday's law), which drives the current.

The current is the same in both cases – it depends only on the *relative* motion – yet the 19th-century explanations were completely different. Einstein (1905) concluded that the distinction between $\mathbf{E}$ and $\mathbf{B}$ must depend on the observer. This is the central idea of Unit 2.

1.2 Charge is invariant

Experiments show that the charge of a particle does not depend on its speed: a hydrogen molecule and a helium atom (whose electrons move very differently relative to their nuclei) are both exactly neutral. Unlike energy or length, **charge is a Lorentz scalar**. Charge *density* is not invariant, however: a moving charge distribution is length-contracted, so $\rho = \gamma\rho_0$.

2 Interdependence of electric and magnetic fields

One observer's electric field is another observer's magnetic field.

- A charge at rest produces only $\mathbf{E}$. Seen from a moving frame it is a moving charge – a current – and produces both $\mathbf{E}$ and $\mathbf{B}$.

- A neutral current-carrying wire produces only **B** in the lab. Seen from a frame moving along the wire, the wire becomes *charged* (Section 3) and produces an **E** as well.
- **E** and **B** are therefore not separate entities; they are two aspects of a single **electromagnetic field**. When we change frames, their components mix – just as x and t mix under a Lorentz transformation.

Griffiths summarises: given electrostatics (Coulomb's law) and special relativity, *magnetism must exist.* Magnetism is a relativistic effect of electric charges in motion.

3 Force and fields near a current-carrying conductor

Model a neutral wire carrying current (Griffiths §12.3.1, adapted from Purcell): a line of positive charge $+\lambda$ moving to the right at speed v and a line of negative charge $-\lambda$ moving to the left at speed v. The net charge is zero and the current is

$$I = 2\lambda v.$$

A point charge q at distance s moves to the right with speed u (Fig. 2a). In this frame S the wire is neutral, so there is *no electric force* on q.

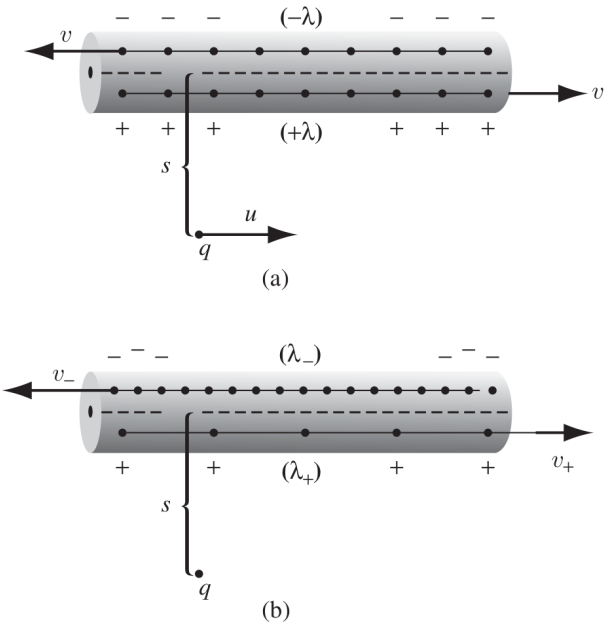


Figure 2: (a) In frame S positive charges move right and negative charges move left with the same speed v: the wire is neutral; the charge q moves with speed u. (b) In the rest frame $\bar{S}$ of q, the two lines move with different speeds $v_\pm$, are Lorentz-contracted by different amounts, and the wire carries a net charge.
Source: D. J. Griffiths, Introduction to Electrodynamics*, Fig. 12.34*

Derivation [5 marks]: The magnetic force on a charge near a wire, from relativity

(i) **Go to the rest frame $\bar{S}$ of q** (moving right at u). By velocity addition the line charges move at $v_\pm = (v \mp u)/(1 \mp vu/c^2)$. Since $v_- > v_+$, the negative line is more contracted, so its charge density is larger in magnitude: in $\bar{S}$ *the wire is negatively charged.*

(ii) **Net line charge in $\bar{S}$** (the algebra with $\gamma_\pm$ is skipped; Griffiths Eq. 12.83):

$$\lambda_{\text{tot}} = -\frac{2\lambda u v}{c^2\sqrt{1-u^2/c^2}}.$$

(iii) **Electric force in $\bar{S}$.** A line charge gives $E = \lambda_{\text{tot}}/(2\pi\varepsilon_0 s)$, so the force on the (stationary) charge q is

$$\bar{F} = qE = -\frac{\lambda v}{\pi\varepsilon_0 c^2 s}\,\frac{qu}{\sqrt{1-u^2/c^2}} \quad \text{(attractive)}.$$

(iv) **Back to S.** A force perpendicular to the motion, on a particle at rest in $\bar{S}$, transforms as $F = \bar{F}\sqrt{1-u^2/c^2}$ (Unit 1, four-force). The square roots cancel:

$$F = -\frac{\lambda v}{\pi\varepsilon_0 c^2}\,\frac{qu}{s}.$$

(v) Using $c^2 = 1/(\varepsilon_0\mu_0)$ and $I = 2\lambda v$:

$$F = -qu\Big(\frac{\mu_0 I}{2\pi s}\Big) = -quB.$$

The bracket is exactly the magnetic field of a long straight wire, and $F = quB$ is the magnetic Lorentz force. ■

Conclusion.
The force on q is *purely electric* in $\bar{S}$ (where the wire is charged and q is at rest) and *purely magnetic* in S (where the wire is neutral and q moves). Electrostatics + relativity ⇒ magnetism. Fields near the wire: in S, $\mathbf{E} = 0$, $B = \mu_0 I/2\pi s$ (circling the wire); in $\bar{S}$, both $\mathbf{E}$ (radial) and $\mathbf{B}$ are present.

How can such a tiny effect be so big?.
Electrons drift in a copper wire at only $\sim 10^{-4}\,\mathrm{m\,s^{-1}}$, so $v^2/c^2 \sim 10^{-25}$! Yet magnetic forces between wires are easily felt. The reason: a wire contains an enormous amount of charge ($\sim 10^4\,\mathrm{C}$ per metre in each sign). Electric forces are normally cancelled almost perfectly; the tiny relativistic imbalance of an enormous charge is what we call magnetism.

Solved Example 1: Drift velocity and the magnetic force
(a) A copper wire of cross-section $1\,\mathrm{mm}^2$ carries $I = 1\,\mathrm{A}$; free-electron density $n = 8.5\times10^{28}\,\mathrm{m^{-3}}$. Drift speed: $v_d = I/(neA) = 1/(8.5\times10^{28}\times1.6\times10^{-19}\times10^{-6}) = 7.4\times10^{-5}\,\mathrm{m\,s^{-1}}$.
(b) An electron moves parallel to a long wire carrying $10\,\mathrm{A}$, at distance $1\,\mathrm{cm}$, with speed $2\times10^6\,\mathrm{m\,s^{-1}}$. $B = \mu_0 I/2\pi s = 2\times10^{-4}\,\mathrm{T}$, $F = quB = (1.6\times10^{-19})(2\times10^6)(2\times10^{-4}) = 6.4\times10^{-17}\,\mathrm{N}$. In the electron's rest frame the same force is electric, due to a net charge on the wire.

4 Lorentz transformation of electric and magnetic fields

E and **B** are not the spatial parts of four-vectors; their transformation is found by studying simple charge configurations (Griffiths §12.3.2). We assume (i) charge is invariant and (ii) fields transform in the same way however they are produced.

4.1 Perpendicular components of E: the parallel-plate capacitor

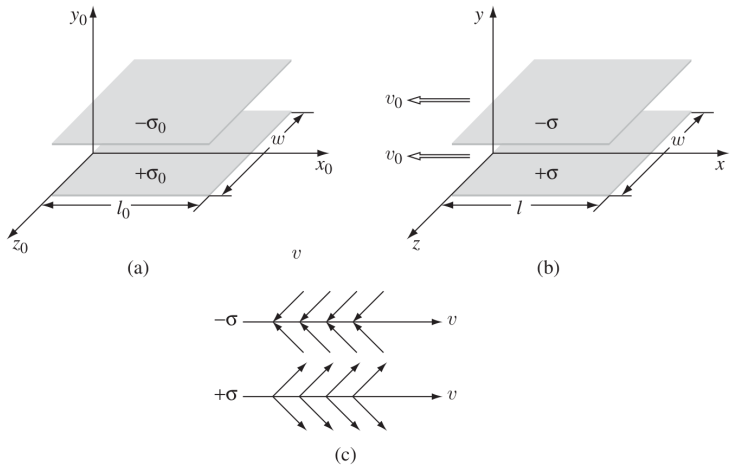


Figure 3: (a) A capacitor at rest in S_0 with surface charges $\pm\sigma_0$. (b) The same capacitor seen from S, moving to the right at v_0; the plates move left and are contracted along x.
Source: D. J. Griffiths, Introduction to Electrodynamics*, Fig. 12.35*

Derivation [5 marks]: Transformation of $E_\perp$ and $E_\parallel$

$E_\perp$**:** In S_0 (capacitor at rest, plates parallel to the xz-plane) the field between the plates is $E_0 = \sigma_0/\varepsilon_0$ along y. In S, moving along x, the total charge on a plate is the same (charge invariance) and the width w is unchanged, but the length l is contracted, $l = l_0/\gamma_0$. So the surface charge density increases: $\sigma = \gamma_0\sigma_0$. Gauss's law still gives $E = \sigma/\varepsilon_0$, hence

$$E^\perp = \gamma_0 E_0^\perp.$$

$E_\parallel$**:** Turn the capacitor so its plates are parallel to the yz-plane (Fig. 4). Now only the plate separation d is contracted; l, w and σ are unchanged, and $E = \sigma/\varepsilon_0$ does not depend on d. Hence

$$E^\parallel = E_0^\parallel. \qquad \blacksquare$$

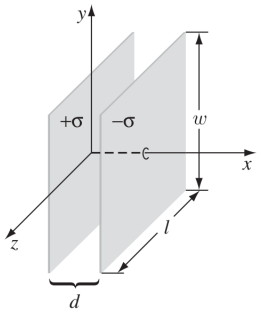


Figure 4: Capacitor with plates parallel to the yz-plane: only the plate separation d is contracted, so the parallel component of **E** is unchanged.
Source: D. J. Griffiths, Introduction to Electrodynamics*, Fig. 12.36*

4.2 The general rules

In frame S the moving capacitor plates are also surface currents, so there is a magnetic field $B_z = -\mu_0\sigma v_0$ as well. Repeating the argument from a third frame $\bar{S}$ (we skip the algebra; Griffiths

Eqs. 12.98–12.105) gives the transformation of E_y and B_z together. A long solenoid along x (Fig. 5) shows that B_x is unchanged: the number of turns per unit length increases by γ (contraction) but the current decreases by γ (time dilation).

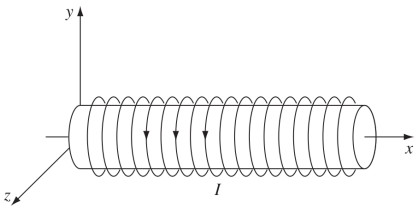


Figure 5: A long solenoid along the x-axis at rest in S. In $\bar{S}$ the turns are closer together but the current is smaller; the two factors of γ cancel, so $\bar{B}_x = B_x$.
Source: D. J. Griffiths, Introduction to Electrodynamics*, Fig. 12.40*

Transformation of the fields ($\bar{S}$ moves with speed v along x).

$$\bar{E}_x = E_x, \quad \bar{E}_y = \gamma\,(E_y - vB_z), \qquad \bar{E}_z = \gamma\,(E_z + vB_y),$$
$$\bar{B}_x = B_x, \quad \bar{B}_y = \gamma\Big(B_y + \frac{v}{c^2}E_z\Big), \quad \bar{B}_z = \gamma\Big(B_z - \frac{v}{c^2}E_y\Big).$$

In words: components *parallel* to the motion are unchanged; *perpendicular* components mix, $\bar{\mathbf{E}}_\perp = \gamma(\mathbf{E} + \mathbf{v}\times\mathbf{B})_\perp$, $\bar{\mathbf{B}}_\perp = \gamma(\mathbf{B} - \mathbf{v}\times\mathbf{E}/c^2)_\perp$.

Two special cases (Griffiths Eq. 12.110, 12.111):

- If $\mathbf{B} = 0$ in S: $\bar{\mathbf{B}} = -\dfrac{1}{c^2}(\mathbf{v}\times\bar{\mathbf{E}})$.
- If $\mathbf{E} = 0$ in S: $\bar{\mathbf{E}} = \mathbf{v}\times\bar{\mathbf{B}}$.

If either field vanishes in one frame, the fields in any other frame are simply related.
Low-speed limit ($\gamma \approx 1$): $\bar{\mathbf{E}} \approx \mathbf{E} + \mathbf{v}\times\mathbf{B}$. This is why the moving loop and the moving magnet give the same emf: the "magnetic force" $q\mathbf{v}\times\mathbf{B}$ in one frame is the "electric force" $q\bar{\mathbf{E}}$ in the other.

Field invariants.
Although $\mathbf{E}$ and $\mathbf{B}$ change, two combinations are the same in all frames:

$$\mathbf{E}\cdot\mathbf{B} = \bar{\mathbf{E}}\cdot\bar{\mathbf{B}}, \qquad E^2 - c^2B^2 = \bar{E}^2 - c^2\bar{B}^2.$$

Consequences: if $\mathbf{E}\perp\mathbf{B}$ in one frame, they are perpendicular in all frames; if $E > cB$ in one frame, $\mathbf{B}$ can be transformed away but $\mathbf{E}$ never can (and vice versa); a pure electromagnetic wave ($E = cB$, $\mathbf{E}\perp\mathbf{B}$) looks like a wave to everyone.

Derivation [5 marks]: $\mathbf{E}\cdot\mathbf{B}$ is invariant (for a boost along x)

$$\bar{\mathbf{E}}\cdot\bar{\mathbf{B}} = E_xB_x + \gamma^2\Big[(E_y - vB_z)\Big(B_y + \tfrac{v}{c^2}E_z\Big) + (E_z + vB_y)\Big(B_z - \tfrac{v}{c^2}E_y\Big)\Big].$$

Expanding the bracket: $E_yB_y + E_zB_z + \frac{v}{c^2}(E_yE_z - E_zE_y) + v(B_yB_z - B_zB_y) - \frac{v^2}{c^2}(E_zB_z + E_yB_y) = (1-\beta^2)(E_yB_y + E_zB_z)$. Since $\gamma^2(1-\beta^2) = 1$, $\bar{\mathbf{E}}\cdot\bar{\mathbf{B}} = E_xB_x + E_yB_y + E_zB_z = \mathbf{E}\cdot\mathbf{B}$. ■

Solved Example 2: Transforming a pure electric field

In S: $\mathbf{E} = (0,\ 1.0\times10^5,\ 0)\,\mathrm{V/m}$, $\mathbf{B} = 0$. Find the fields in $\bar{S}$ moving at $v = 0.6c$ along x ($\gamma = 1.25$).

$$\bar{E}_y = \gamma E_y = 1.25\times10^5\ \mathrm{V/m}, \qquad \bar{B}_z = -\gamma\frac{v}{c^2}E_y = -\frac{1.25\times1.8\times10^8\times10^5}{9\times10^{16}} = -2.5\times10^{-4}\ \mathrm{T}.$$

Check: $E^2 - c^2B^2 = 10^{10}$ in S; in $\bar{S}$, $1.5625\times10^{10} - 9\times10^{16}\times6.25\times10^{-8} = 10^{10}$ ✓.

Solved Example 3: Crossed fields: transforming the electric field away

In S: $\mathbf{E} = 3.0\times10^4\,\hat{\mathbf{y}}\,\mathrm{V/m}$ and $\mathbf{B} = 2.0\times10^{-4}\,\hat{\mathbf{z}}\,\mathrm{T}$. Here $cB = 6\times10^4\,\mathrm{V/m} > E$ and $\mathbf{E}\cdot\mathbf{B} = 0$, so a frame exists with no electric field. Require $\bar{E}_y = \gamma(E_y - vB_z) = 0$: $v = E/B = 1.5\times10^8\,\mathrm{m/s} = 0.5c$, $\gamma = 1.155$.

$$\bar{B}_z = \gamma\Big(B_z - \frac{v}{c^2}E_y\Big) = 1.155\,(2.0 - 0.5)\times10^{-4} = 1.73\times10^{-4}\ \mathrm{T}.$$

Check: $E^2 - c^2B^2 = 9\times10^8 - 3.6\times10^9 = -2.7\times10^9$; $-c^2\bar{B}^2 = -9\times10^{16}\times3.0\times10^{-8} = -2.7\times10^9$ ✓. (This is the principle of the $\mathbf{E}\times\mathbf{B}$ drift and of velocity selectors.)

5 The field of a uniformly moving point charge

We want the fields of a charge q moving with constant velocity $\mathbf{v}$ in the lab. In the charge's own rest frame S_0 there is only the Coulomb field. Griffiths (Example 12.14) obtains its field in a few lines from the transformation rules: in S_0 the field is pure Coulomb; transform $\mathbf{E}$ (the perpendicular components gain a factor γ) and express the coordinates of the field point in the lab frame (the x-coordinate also gains a factor γ). We quote the result.

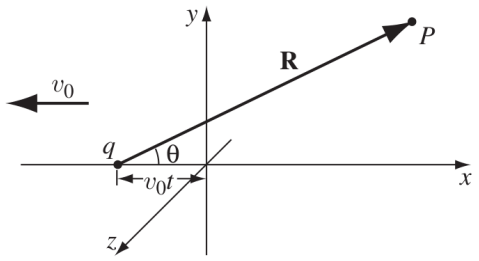


Figure 6: Geometry for the field of a uniformly moving charge: $\mathbf{R}$ is the vector from the *present* position of q to the field point P, and θ is the angle between $\mathbf{R}$ and the velocity.
Source: D. J. Griffiths, Introduction to Electrodynamics, *Fig. 12.37*

Fields of a point charge moving with constant velocity v.

$$\mathbf{E} = \frac{q}{4\pi\varepsilon_0}\,\frac{1 - v^2/c^2}{\left[1 - (v^2/c^2)\sin^2\theta\right]^{3/2}}\,\frac{\hat{\mathbf{R}}}{R^2}, \qquad \mathbf{B} = \frac{1}{c^2}\,(\mathbf{v}\times\mathbf{E}).$$

- $\mathbf{E}$ points radially from the *present* position of the charge (not the retarded one).
- Along the line of motion ($\theta = 0, \pi$): $E = \dfrac{1}{\gamma^2}\dfrac{q}{4\pi\varepsilon_0 R^2}$ (weaker).
- Perpendicular to the motion ($\theta = \pi/2$): $E = \gamma\,\dfrac{q}{4\pi\varepsilon_0 R^2}$ (stronger).

- The field lines crowd into the plane perpendicular to the motion – the "pancake" field.
- For $v \ll c$, $\mathbf{B} \approx \dfrac{\mu_0}{4\pi}\dfrac{q\,\mathbf{v}\times\hat{\mathbf{R}}}{R^2}$ – the Biot–Savart law for a point charge.

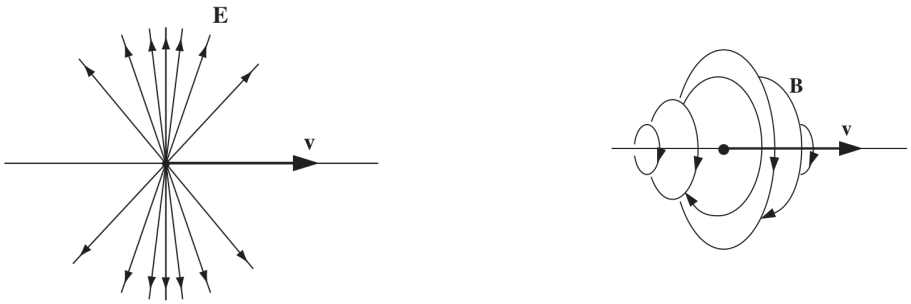


Figure 7: **Left:** electric field lines of a fast-moving charge are concentrated in the plane perpendicular to its velocity. **Right:** magnetic field lines circle the line of motion.
Source: D. J. Griffiths, Introduction to Electrodynamics*, Figs. 10.10 and 10.11*

Solved Example 4: A fast proton

A proton moves at $0.99c$ ($\gamma = 7.09$). At $R = 1\,\text{nm}$ the Coulomb field would be $E_0 = kq/R^2 = (8.99\times10^9)(1.6\times10^{-19})/10^{-18} = 1.44\times10^9\,\text{V/m}$.

- Perpendicular to the motion: $E = \gamma E_0 = 1.02\times10^{10}\,\text{V/m}$, and $B = vE/c^2 = (0.99)(1.02\times10^{10})/(3\times10^8) = 34\,\text{T}$.
- Straight ahead: $E = E_0/\gamma^2 = 1.44\times10^9/50.3 = 2.9\times10^7\,\text{V/m}$.

6 Force between moving charges

Derivation [5 marks]: Two charges moving side by side

Two equal charges q move with the same velocity $v\hat{\mathbf{x}}$, separated by d along y.

Rest frame: pure Coulomb repulsion $F_0 = \dfrac{q^2}{4\pi\varepsilon_0 d^2}$.

Lab frame, using fields: the field of one charge at the other ($\theta = 90°$) is $E_y = \gamma q/(4\pi\varepsilon_0 d^2)$ and $B_z = vE_y/c^2$. The force on the second charge is

$$F_y = q(E_y - vB_z) = qE_y\Big(1 - \frac{v^2}{c^2}\Big) = \frac{\gamma q^2}{4\pi\varepsilon_0 d^2}\cdot\frac{1}{\gamma^2} = \frac{F_0}{\gamma}.$$

The electric repulsion is increased by γ, but the magnetic attraction (parallel currents) more than compensates.

Lab frame, transforming the force: a perpendicular force on a particle at rest in S_0 transforms as $F = F_0/\gamma$. Same answer. ■

This resolves the puzzle raised in Unit 1, Part A: the force *is* different in different frames, exactly as relativistic dynamics requires, and the magnetic force is what makes the two descriptions agree.

It is also why intense parallel electron beams in accelerators spread out much less than Coulomb's law alone would suggest.

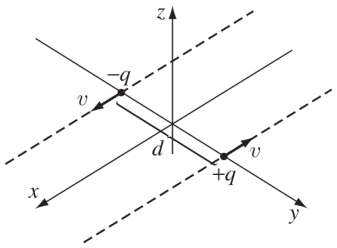


Figure 8: Two charges $\pm q$ on parallel trajectories a distance d apart, moving in opposite directions (Griffiths Problem 12.46). For opposite velocities the magnetic force adds to the electric attraction: $F = \gamma(1+\beta^2)\, q^2/(4\pi\varepsilon_0 d^2)$.
Source: D. J. Griffiths, Introduction to Electrodynamics, *Fig. 12.42*

Solved Example 5: Two electrons in a beam

Two electrons move side by side at $0.8c$ ($\gamma = 5/3$), 1.0 mm apart. $F_0 = \dfrac{(8.99\times10^{9})(1.6\times10^{-19})^2}{(10^{-3})^2} = 2.3\times10^{-22}\,\mathrm{N}$; in the lab $F = F_0/\gamma = 1.4\times10^{-22}\,\mathrm{N}$ (still repulsive, but 40% weaker).

7 Tensor form: invariance and covariance of Maxwell's equations

7.1 Invariance versus covariance

Two words to keep apart.

- A quantity is **invariant** if it has the same *value* in all inertial frames: c, charge q, rest mass, the interval s^2, $\mathbf{E}\cdot\mathbf{B}$, $E^2 - c^2B^2$.
- An equation is **covariant** if it has the same *form* in all inertial frames, although the quantities in it change. An equation of the form "tensor = tensor" is automatically covariant.

Maxwell's equations are *covariant* under Lorentz transformations (not under Galilean ones – Unit 1, Part A). Writing them in tensor form makes this obvious.

7.2 Recap of four-vector notation

$x^\mu = (ct, x, y, z)$; $x'^\mu = \Lambda^\mu{}_\nu x^\nu$ (summation over repeated indices); metric $g_{\mu\nu} = \mathrm{diag}(-1,1,1,1)$; lowering an index: $a_\mu = g_{\mu\nu}a^\nu = (-a^0, \mathbf{a})$; invariant $a\cdot b = a_\mu b^\mu$. A rank-2 tensor transforms as $\bar{T}^{\mu\nu} = \Lambda^\mu{}_\lambda \Lambda^\nu{}_\sigma T^{\lambda\sigma}$. The four-gradient $\partial_\mu = \partial/\partial x^\mu = \left(\frac{1}{c}\frac{\partial}{\partial t}, \nabla\right)$.

7.3 The field tensor

E and **B** together have six components and mix under boosts. A rank-2 *antisymmetric* tensor in four dimensions has exactly six independent components. Griffiths (§12.3.3) shows that

Electromagnetic field tensor and its dual.

$$F^{\mu\nu} = \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & B_z & -B_y \\ -E_y/c & -B_z & 0 & B_x \\ -E_z/c & B_y & -B_x & 0 \end{pmatrix}, \qquad G^{\mu\nu} = \begin{pmatrix} 0 & B_x & B_y & B_z \\ -B_x & 0 & -E_z/c & E_y/c \\ -B_y & E_z/c & 0 & -E_x/c \\ -B_z & -E_y/c & E_x/c & 0 \end{pmatrix}.$$

Transforming $F^{\mu\nu}$ with $\bar{F}^{\mu\nu} = \Lambda^{\mu}{}_{\lambda}\Lambda^{\nu}{}_{\sigma}F^{\lambda\sigma}$ reproduces exactly the field-transformation rules of Section 4. ($G^{\mu\nu}$ is obtained from $F^{\mu\nu}$ by $\mathbf{E}/c \to \mathbf{B}$, $\mathbf{B} \to -\mathbf{E}/c$.) The invariants are the scalars $F_{\mu\nu}F^{\mu\nu} = 2(B^2 - E^2/c^2)$ and $F_{\mu\nu}G^{\mu\nu} = -4(\mathbf{E}\cdot\mathbf{B})/c$.

Solved Example 6: Reading off a component

Check one entry: $\bar{F}^{01} = \Lambda^{0}{}_{\lambda}\Lambda^{1}{}_{\sigma}F^{\lambda\sigma}$. With $\Lambda^{0}{}_{0} = \Lambda^{1}{}_{1} = \gamma$, $\Lambda^{0}{}_{1} = \Lambda^{1}{}_{0} = -\gamma\beta$, only $(\lambda,\sigma) = (0,1),(1,0)$ contribute ($F^{00} = F^{11} = 0$): $\bar{F}^{01} = \gamma^2 F^{01} + \gamma^2\beta^2 F^{10} = \gamma^2(1-\beta^2)F^{01} = F^{01}$. So $\bar{E}_x = E_x$, as found with the capacitor.

7.4 The current density four-vector and charge conservation

Charge density and current density combine into a four-vector (Griffiths §12.3.4):

$$J^{\mu} = (c\rho,\ J_x,\ J_y,\ J_z) = \rho_0\,\eta^{\mu},$$

where ρ_0 is the proper charge density and $\eta^{\mu} = \gamma(c, \mathbf{u})$ the proper (four-)velocity.

Derivation [5 marks]: Continuity equation in covariant form

$$\frac{\partial J^{\mu}}{\partial x^{\mu}} = \frac{\partial(c\rho)}{\partial(ct)} + \frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} + \frac{\partial J_z}{\partial z} = \frac{\partial \rho}{\partial t} + \nabla\cdot\mathbf{J}.$$

Local conservation of charge, $\partial\rho/\partial t + \nabla\cdot\mathbf{J} = 0$, is therefore $\partial_{\mu}J^{\mu} = 0$: the four-divergence of a four-vector, a Lorentz scalar. If it vanishes in one frame it vanishes in all. ■

7.5 Maxwell's equations in tensor form

Maxwell's equations (Griffiths Eq. 12.127).

$$\frac{\partial F^{\mu\nu}}{\partial x^{\nu}} = \mu_0 J^{\mu} \qquad \text{(Gauss's law and Ampère–Maxwell law)},$$

$$\frac{\partial G^{\mu\nu}}{\partial x^{\nu}} = 0 \qquad \text{(no magnetic monopoles and Faraday's law)}.$$

Both sides are four-vectors, so the equations have the same form in every inertial frame: Maxwell's equations are Lorentz-covariant.

Derivation [5 marks]: Recovering Gauss's law and the Ampère–Maxwell law

$\mu = 0$**:** $\dfrac{\partial F^{0\nu}}{\partial x^{\nu}} = \dfrac{\partial F^{01}}{\partial x} + \dfrac{\partial F^{02}}{\partial y} + \dfrac{\partial F^{03}}{\partial z} = \dfrac{1}{c}\nabla\cdot\mathbf{E} = \mu_0 J^0 = \mu_0 c\rho$, so $\nabla\cdot\mathbf{E} = \mu_0 c^2\rho = \rho/\varepsilon_0$.

$\mu = 1$**:** $\dfrac{\partial F^{10}}{\partial(ct)} + \dfrac{\partial F^{12}}{\partial y} + \dfrac{\partial F^{13}}{\partial z} = -\dfrac{1}{c^2}\dfrac{\partial E_x}{\partial t} + \dfrac{\partial B_z}{\partial y} - \dfrac{\partial B_y}{\partial z} = \mu_0 J_x$, i.e. $(\nabla\times\mathbf{B})_x = \mu_0 J_x + \mu_0\varepsilon_0\,\partial E_x/\partial t$.

With $\mu = 2, 3$ this is the Ampère–Maxwell law. (The $G^{\mu\nu}$ equation gives $\nabla\cdot\mathbf{B} = 0$ and Faraday's law in the same way.) ■

7.6 The Lorentz force in tensor form

The Minkowski (four-)force on a charge q with proper velocity η^ν is

$$K^\mu = q\,\eta_\nu F^{\mu\nu},$$

whose spatial part is $\mathbf{K} = \gamma\, q(\mathbf{E} + \mathbf{u}\times\mathbf{B})$ – i.e. the ordinary force $\mathbf{F} = \mathbf{K}/\gamma = q(\mathbf{E} + \mathbf{u}\times\mathbf{B})$, the Lorentz force law, valid at all speeds.

7.7 Relativistic potentials

The scalar and vector potentials form a four-vector $A^\mu = (V/c,\ \mathbf{A})$. Then

$$F^{\mu\nu} = \frac{\partial A^\nu}{\partial x_\mu} - \frac{\partial A^\mu}{\partial x_\nu}$$

(which reproduces $\mathbf{E} = -\nabla V - \partial\mathbf{A}/\partial t$, $\mathbf{B} = \nabla\times\mathbf{A}$). In the Lorenz gauge $\partial_\mu A^\mu = \dfrac{1}{c^2}\dfrac{\partial V}{\partial t} + \nabla\cdot\mathbf{A} = 0$, Maxwell's equations become the single covariant wave equation

$$\Box^2 A^\mu = -\mu_0 J^\mu, \qquad \Box^2 \equiv \partial_\mu\partial^\mu = \nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}.$$

Derivation [5 marks]: The d'Alembertian $\Box^2$ is Lorentz invariant
With $x' = \gamma(x - vt)$, $t' = \gamma(t - vx/c^2)$ the chain rule gives

$$\frac{\partial}{\partial x} = \gamma\Big(\frac{\partial}{\partial x'} - \frac{v}{c^2}\frac{\partial}{\partial t'}\Big), \qquad \frac{\partial}{\partial t} = \gamma\Big(\frac{\partial}{\partial t'} - v\frac{\partial}{\partial x'}\Big).$$

Squaring,

$$\frac{\partial^2}{\partial x^2} - \frac{1}{c^2}\frac{\partial^2}{\partial t^2} = \gamma^2\Big[\Big(1 - \frac{v^2}{c^2}\Big)\frac{\partial^2}{\partial x'^2} - \frac{1}{c^2}\Big(1 - \frac{v^2}{c^2}\Big)\frac{\partial^2}{\partial t'^2}\Big] = \frac{\partial^2}{\partial x'^2} - \frac{1}{c^2}\frac{\partial^2}{\partial t'^2},$$

the cross terms $\pm 2\gamma^2(v/c^2)\,\partial^2/\partial x'\partial t'$ cancelling. With $\partial/\partial y = \partial/\partial y'$, $\partial/\partial z = \partial/\partial z'$: $\Box^2 = \Box'^2$. Hence the wave equation (and the speed of light) is the same in all frames – compare the Galilean result in Unit 1, Part A. ■

8 Applications of relativistic electrodynamics

8.1 Charged particles in a magnetic field: cyclotrons and synchrotrons

In a uniform magnetic field the force $q\mathbf{u}\times\mathbf{B}$ is perpendicular to $\mathbf{u}$, so $|\mathbf{u}|$, γ and E stay constant. $\mathrm{d}\mathbf{p}/\mathrm{d}t = q\mathbf{u}\times\mathbf{B}$ with $\mathbf{p} = \gamma m\mathbf{u}$ gives a circle with

$$p = \gamma m u = qBR, \qquad \omega = \frac{u}{R} = \frac{qB}{\gamma m}.$$

- The radius measures the *relativistic* momentum: $R = p/(qB)$.
- The revolution frequency *decreases* as γ grows. A fixed-frequency cyclotron therefore falls out of step at high energy; synchrocyclotrons vary the frequency and synchrotrons increase B as p grows.

Solved Example 7: Bending in a magnetic field

(a) An electron with $T = 1\,\text{MeV}$ ($pc = 1.42\,\text{MeV}$, Unit 1) in $B = 0.01\,\text{T}$:

$$R = \frac{p}{eB} = \frac{1.42 \times 10^6\,\text{eV}/c}{e\,(0.01\,\text{T})} = \frac{1.42 \times 10^6}{(3 \times 10^8)(0.01)}\,\text{m} = 0.47\ \text{m}.$$

(b) LHC protons with $E = 6.8\,\text{TeV}$ ($pc \approx 6.8\,\text{TeV}$) in dipole magnets of $8.33\,\text{T}$:

$$R = \frac{6.8 \times 10^{12}}{(3 \times 10^8)(8.33)}\,\text{m} \approx 2.7\ \text{km}.$$

Here $\gamma = E/mc^2 \approx 7250$.



Figure 9: The tunnel of the Large Hadron Collider at CERN. Superconducting dipole magnets (blue) bend protons of $\gamma \approx 7000$ around a 27 km ring; the radius is fixed by $p = qBR$.
Source: Wikimedia Commons, "Views of the LHC tunnel sector 3-4, tirage 1.jpg" (M. Brice, CERN; CC BY-SA 3.0)

8.2 Acceleration by an electric field

A charge accelerated from rest through potential difference V gains kinetic energy $T = qV$ (the work–energy theorem holds relativistically): $\gamma = 1 + qV/mc^2$. Electrons reach $\gamma = 2$ after only 511 kV; their speed then hardly increases although their energy keeps growing.

8.3 Synchrotron radiation and relativistic beaming

A charge on a curved path radiates. For $\gamma \gg 1$ the radiation is concentrated in a narrow forward cone of half-angle $\sim 1/\gamma$ (the headlight effect of Unit 1). Synchrotron light sources exploit this to make intense X-ray beams; in astronomy, synchrotron radiation from relativistic electrons spiralling in magnetic fields is the radio emission of radio galaxies and quasars (Unit 3).

8.4 An electromagnetic wave seen from a moving frame

A plane wave $\mathbf{E} = E_0 \cos(kx - \omega t)\,\hat{\mathbf{y}}$, $\mathbf{B} = (E_0/c)\cos(kx - \omega t)\,\hat{\mathbf{z}}$ viewed from $\bar{S}$ moving along x with speed v:

$$\bar{E}_y = \gamma(E_y - vB_z) = \gamma(1-\beta)E_y = \sqrt{\frac{1-\beta}{1+\beta}}\,E_y, \qquad \bar{B}_z = \bar{E}_y/c.$$

The amplitude is reduced by the same Doppler factor as the frequency, and the intensity by its square. As $v \to c$ the wave fades away – the answer to the teenage Einstein's question of what one would see riding alongside a light beam: nothing. (Griffiths Problem 12.48.)

8.5 Everyday magnetism

Electric motors, transformers and the attraction between parallel wires are relativistic effects at drift speeds of a fraction of a millimetre per second (Section 3).

Summary of Unit 2

Formula sheet.

Force near wire	In charge's frame: wire charged, force electric; in lab: $F = quB$, $B = \mu_0 I/2\pi s$
E, B transformation	$\bar{E}_\parallel = E_\parallel$, $\bar{B}_\parallel = B_\parallel$; $\bar{\mathbf{E}}_\perp = \gamma(\mathbf{E} + \mathbf{v}\times\mathbf{B})_\perp$, $\bar{\mathbf{B}}_\perp = \gamma(\mathbf{B} - \mathbf{v}\times\mathbf{E}/c^2)_\perp$
Special cases	$\mathbf{B} = 0 \Rightarrow \bar{\mathbf{B}} = -\mathbf{v}\times\bar{\mathbf{E}}/c^2$; $\mathbf{E} = 0 \Rightarrow \bar{\mathbf{E}} = \mathbf{v}\times\bar{\mathbf{B}}$
Invariants	$\mathbf{E}\cdot\mathbf{B}$, $E^2 - c^2B^2$
Moving charge	$\mathbf{E} = \dfrac{q}{4\pi\varepsilon_0}\dfrac{1-\beta^2}{(1-\beta^2\sin^2\theta)^{3/2}}\dfrac{\hat{\mathbf{R}}}{R^2}$, $\mathbf{B} = \mathbf{v}\times\mathbf{E}/c^2$
Parallel moving charges	$F = F_0/\gamma$
Four-current	$J^\mu = (c\rho, \mathbf{J})$, $\partial_\mu J^\mu = 0$
Maxwell	$\partial_\nu F^{\mu\nu} = \mu_0 J^\mu$, $\partial_\nu G^{\mu\nu} = 0$
Potentials	$A^\mu = (V/c, \mathbf{A})$, $\Box^2 A^\mu = -\mu_0 J^\mu$ (Lorenz gauge)
Circular motion	$p = qBR$, $\omega = qB/\gamma m$